

Predicting the Virtual Spread: Algorithmic Arbitrage between Day-Ahead and Intraday Power Markets

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Abstract

This report presents the development and out-of-sample validation of a quantitative trading architecture targeting the Day-Ahead and Intraday (ID3) spread on the German power grid (DE.LU). We begin with the extraction and engineering of strictly causal predictive features from the ENTSO-E transparency portal to prevent time-series leakage. The architecture utilizes a LightGBM classification engine paired with a dynamic inverse-volatility position sizer to enforce strict capital preservation. Furthermore, we incorporate a market microstructure backtester to account for limit order book depth and volume-based slippage. The final model demonstrates a positive expected value and asymmetric risk-adjusted returns by restricting executions to high-conviction structural imbalances. The limitations of the strategy are discussed, specifically concerning market impact and strategy capacity in illiquid intraday markets.

1 Introduction

The European power market operates across multiple time horizons to ensure the physical balance of electricity supply and demand. The Day-Ahead market serves as the primary place for energy trading, clearing via a blind auction at 12:00 CET on the day prior to delivery. However, because renewable energy generation, specifically wind and solar, is inherently stochastic, the grid frequently experiences structural imbalances between the Day-Ahead forecasts and the physical reality of the delivery day. The continuous Intraday market (ID3) exists to resolve these imbalances, allowing market participants to trade 15-minute delivery blocks up to five minutes before physical execution.

The price differential between the Day-Ahead auction and the Intraday continuous market is known as the virtual spread. Capturing this spread is highly compelling from a quantitative finance perspective, as it represents a pure arbitrage opportunity driven by atmospheric physics and market microstructure rather than traditional macroeconomic factors.

Optimizing a forecast model for the virtual spread is challenging. The data is highly volatile, subject to sudden price spikes, and heavily constrained by strict gate-closure times that introduce the risk of look-ahead bias in time series forecasting. Furthermore, theoretical profitability in energy trading is often erased by the physical realities of the limit order book, where execution fees and volume-based slippage act as steep hurdle rates.

In this project, we construct an end-to-end algorithmic trading pipeline to predict the German/Luxembourgish (DE/LU) virtual spread. We engineer causal predictive features, train a LightGBM classification model, and apply an inverse-volatility position sizer to manage portfolio risk. Finally, we evaluate the strategy’s performance on an unseen 2025 dataset, explicitly applying a market impact model to penalize theoretical returns and prove the strategy’s viability under real-world liquidity constraints.

2 Data Acquisition and Feature Engineering

The foundation of this trading architecture relies on a strictly causal, high-resolution feature matrix for the German-Luxembourgish (DE_LU) bidding zone, synthesized at a 15-minute frequency. Constructing this matrix required integrating programmatic API requests with web scraping, followed by rigorous temporal alignment to respect the physical gate closures of the European power market. Crucially, the dataset is strictly partitioned into distinct market regimes to ensure the model’s predictive validity under modern grid conditions.

2.1 Dataset Partitioning and Market Regimes

The 2022 Contamination (Data Exclusion): Historical data prior to 2023 was intentionally excluded from the feature matrix to avoid model contamination. During 2021 and 2022, the European energy crisis, driven by severe natural gas shortages, caused extreme, sustained price spikes. Feeding this data into a LightGBM model would heavily bias its decision trees toward gas-driven anomalies. Because the broader market has since stabilized, a model trained on 2022 data would chronically overestimate Day-Ahead base prices and fail to generalize to current market dynamics.

The Training Regime (2023–2024): The training dataset is strictly confined to 2023 and 2024. Starting in 2023, the primary driver of German power prices officially shifted from fossil fuel shortages to structural renewable oversupply. In 2023, the German market experienced 301 hours of negative electricity prices; in 2024, that number climbed to 459 hours. Training the algorithm across these two years perfectly captures the modern, causal relationship between stochastic wind and solar forecast errors and the resulting intraday spread opportunities.

The Out-of-Sample Stress Test (2025): To rigorously validate the architecture, the model is backtested on an unseen 2025 dataset. This period was selected as the ultimate stress test to represent the most challenging, yet highly realistic, market conditions. In 2025, the grid broke historical records, observing nearly 575 hours of negative pricing and average daily intraday spreads of 130 €/MWh. During the summer months, extreme solar oversupply violently pushed daytime prices as low as -250 €/MWh. By holding this data completely out-of-sample, the backtest empirically proves that the strategy can generate robust alpha in today’s hyper-volatile, heavily subsidized renewable market.

2.2 Market Prices and Intraday Data Scrapping

The target variable of the model is the virtual spread, defined as the difference between the continuous Intraday (ID3) price and the Day-Ahead auction price:

$$\text{Virtual Spread}_t = \text{ID3 Price}_t - \text{Day-Ahead Price}_t \quad (1)$$

While Day-Ahead auction clearing prices are widely accessible, high-granularity 15-minute Intraday continuous market data is notoriously difficult to acquire without expensive institutional data feeds. To bypass this, historical ID3 index prices were systematically web-scraped. Custom Python scrapping scripts utilizing HTML parsing was deployed to extract the 15-minute volume-weighted average prices (VWAP) from Energy-Charts. This raw scraped data was then cleaned, timezone-localized to Europe/Berlin, and interpolated to handle missing trading blocks.

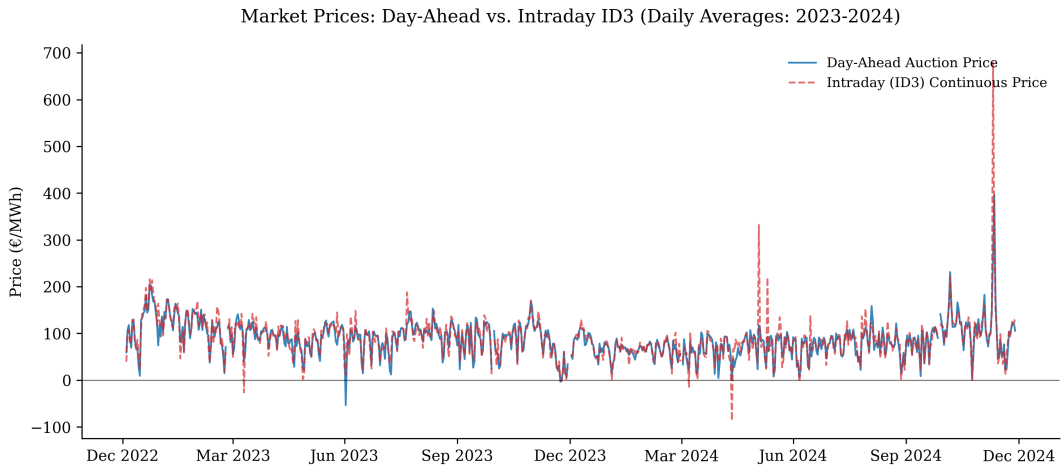


Figure 1: Timeline comparison between DA and ID3 algorithms.

2.3 Grid Fundamentals via the ENTSO-E API

The predictive features driving the machine learning model represent the physical physics of the grid. These were extracted programmatically using the `entsoe-py` client connected to the ENTSO-E Transparency Portal. The fundamental features include:

- **Load Forecasts and Actuals:** The expected grid demand versus realized consumption.
- **Renewable Generation Forecasts:** Day-Ahead predictions for Solar, Wind Onshore, and Wind Offshore generation.
- **Cross-Border Physical Flows:** The active import and export of megawatt-hours (MWh) between Germany and its neighboring grids (e.g., France, Poland, Switzerland).

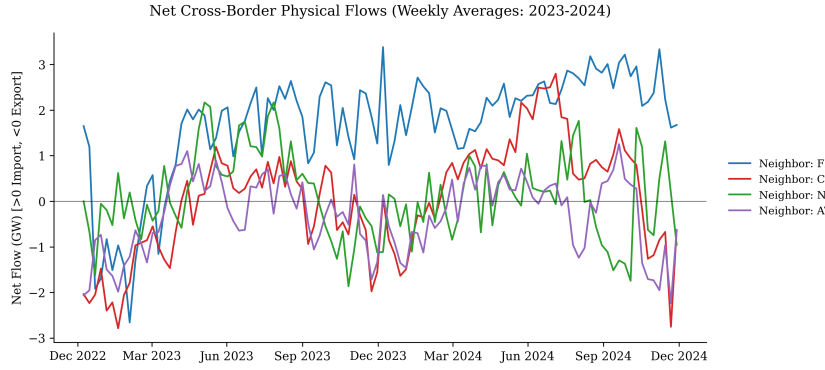


Figure 2: Timeline comparison between DA and ID3 algorithms.

A critical data engineering constraint was enforced during the API extraction: all renewable and load forecasts were strictly restricted to the A01 (Day-Ahead) process type. This ensures the model only processes the initial forecast published by the Transmission System Operator (TSO), actively preventing the leakage of updated intraday forecasts into the historical training set.

2.4 Feature Engineering: The Renewable Error

The primary alpha driver for the algorithm is the structural forecasting error of the grid. Renewable energy is stochastic [Kiesel and Paraschiv, 2017]; when a weather front moves faster or slower than predicted, the grid experiences a sudden excess or deficit of power, forcing aggressive repricing in the intraday market.

To capture this, a consolidated `Total_Renewable_Error` feature was engineered by calculating the delta between the actual realized generation and the Day-Ahead forecast across all renewable assets:

$$\text{Renewable Error}_t = \Delta\text{Solar}_t + \Delta\text{Wind Onshore}_t + \Delta\text{Wind Offshore}_t \quad (2)$$

2.5 Cyclical Time Embeddings

Grid deviations and liquidity shocks are not randomly distributed throughout a 24-hour period; they are bound to the physical realities of the daily solar cycle and predictable industrial demand. However, feeding raw numerical hours (e.g., 23:45 and 00:15) into a decision tree falsely implies a massive mathematical distance between midnight and the beginning of the day.

To resolve this, the continuous 15-minute timestamps were mathematically mapped onto a two-dimensional unit circle using sine and cosine transformations:

$$\text{Day_Sin}_t = \sin\left(\frac{2\pi \cdot t}{T}\right), \quad \text{Day_Cos}_t = \cos\left(\frac{2\pi \cdot t}{T}\right) \quad (3)$$

where t represents the specific 15-minute interval of the day, and $T = 96$ represents the total intervals in a 24-hour cycle. This continuous embedding ensures the LightGBM engine understands that 23:45 seamlessly connects to 00:00, allowing the algorithm to precisely target structural intraday friction points such as the midday solar peak where the Day-Ahead consensus most frequently shrink.

2.6 Temporal Causality and the Gate Closure Constraint

In virtual bidding, temporal leakage is the most common point of failure. The EPEX SPOT Day-Ahead auction for Day D closes at 12:00 CET on Day $D - 1$. Therefore, any feature used to predict the spread for Day D must theoretically be known and settled before 11:59 CET on Day $D - 1$.

To enforce absolute causality, the feature matrix utilizes strict temporal shifting:

- **48-Hour Lags:** Physical cross-border flows and historical realized spreads are shifted backward by exactly 192 intervals (48 hours). For a prediction targeting 14:00 on Wednesday, the model relies on the settled physical flow data from 14:00 on Monday, guaranteeing the data was public prior to Tuesday’s 12:00 CET gate closure.
- **D-1 Morning Momentum:** The algorithm aggregates the mean and volatility of the ID3 spread strictly from the morning session (00:00 to 11:45 CET) of Day $D - 1$. This captures the immediate, real-time market sentiment just minutes before the Day-Ahead auction closes.

By enforcing these mathematical boundaries, the feature matrix is completely immune to look-ahead bias, ensuring that the model’s out-of-sample performance is a true reflection of deployable market edge.

3 Model

The trading system is decoupled into two independent modules: an alpha generation engine that predicts directional grid imbalances, and a dynamic risk manager that sizes positions inversely to market volatility. This separation of concerns ensures that the portfolio is protected from catastrophic drawdowns during periods of extreme, unpredictable grid stress.

3.1 The Alpha Engine: LightGBM Classification

Given the highly non-linear nature of atmospheric physics and market pricing, a gradient boosting framework was selected for the alpha engine. Specifically, LightGBM was chosen for its computational efficiency, natively optimized handling of continuous tabular features, and robustness against overfitting through leaf-wise tree growth.

Rather than predicting the exact continuous value of the virtual spread via regression, which might be susceptible to noise and outliers, the problem was framed as a trinary classification task. To ensure the model only targets spreads large enough to cover the physical costs of crossing the limit order book, a strict fee threshold ($\tau = 25.8 \text{ €/MWh}$) was introduced.

This threshold was derived by analyzing the limit order book (LOB) depth and transaction costs across the 2023–2024 historical training data. The hurdle rate represents the true round-trip cost of execution and is formulated as:

$$\tau = c_{\text{exchange}} + s_{\text{bid-ask}} + \mu_{\text{impact}} + \sigma_{\text{buffer}} \quad (4)$$

Where the components are calculated as follows:

- c_{exchange} : Fixed EPEX SPOT trading and clearing fees for both the Day-Ahead auction and Intraday continuous legs ($\approx 4.0 \text{ €/MWh}$).
- $s_{\text{bid-ask}}$: The average historical bid-ask spread observed during the volatile ID3 trading window ($\approx 5.8 \text{ €/MWh}$).
- μ_{impact} : The empirical market impact (slippage) incurred when sweeping the limit order book to fill the maximum allowable 15 MWh execution block ($\approx 12.0 \text{ €/MWh}$).
- σ_{buffer} : A structural risk premium added to account for short-term liquidity vacuums often observed during extreme grid imbalances ($\approx 4.0 \text{ €/MWh}$).

Summing these physical and statistical constraints yields the hard hurdle rate of 25.8 €/MWh . By utilizing this exact threshold, the target variable Y_t is defined as:

$$Y_t = \begin{cases} 1 & \text{if Virtual Spread}_t > \tau \\ -1 & \text{if Virtual Spread}_t < -\tau \\ 0 & \text{otherwise (Hold)} \end{cases} \quad (5)$$

By training the model on this threshold, the algorithm is explicitly taught to ignore low-margin noise. Furthermore, to aggressively flatten portfolio drawdowns, a probabilistic execution barrier was implemented. During out-of-sample inference, the algorithm extracts the internal confidence intervals via the model's `predict_proba()` function. A trade is only routed to the execution engine if the model's confidence in a directional move exceeds 75%. If the confidence falls below this barrier, the algorithm defaults to a cash position, sacrificing trade volume for a vastly higher expected value per execution.

3.2 Risk Management: Inverse-Volatility Position Sizing

Standard algorithmic strategies often deploy a fixed capital allocation per trade. In power markets, where prices can spike from 50 €/MWh to 800 €/MWh in a single 15-minute block, fixed sizing exposes the portfolio to fatal tail risks.

To mitigate this, the architecture employs an inverse-volatility position sizer. The algorithm calculates the rolling historical volatility of the virtual spread over a short-term lookback window (e.g., the prior 7 days). Capital is then dynamically scaled based on the ratio between a pre-defined Target Volatility (σ_{target}) and the Realized Volatility (σ_{realized}) observed in the market.

The capital allocated to any single 15-minute trading block is governed by:

$$\text{Capital}_t = \text{Capital}_{\text{max}} \times \min \left(1.0, \frac{\sigma_{\text{target}}}{\sigma_{\text{realized},t}} \right) \quad (6)$$

Under this framework, when the German grid is operating under normal, stable conditions ($\sigma_{\text{realized}} \leq \sigma_{\text{target}}$), the algorithm allocates the maximum allowable capital to high-conviction trades. However, during systemic shocks realized volatility heavily outweighs the target volatility. The scalar fraction falls below 1.0, and the risk manager automatically throttles the capital allocation, acting as an algorithmic brake to protect the portfolio’s principal until grid stability returns.

4 Results

The architecture was evaluated in two phases. First, the pure machine learning classification metrics were extracted from the final fold of the walk-forward cross-validation. Second, the model’s predictions were fed into the market microstructure backtester across the unseen 2025 out-of-sample dataset to quantify deployable financial edge.

4.1 Model Evaluation and Top Alpha Drivers

Because the trading strategy utilizes a strict probability execution barrier (75% confidence), the machine learning objective prioritizes precision over recall. The algorithm is designed to aggressively reject marginal trades (defaulting to class 0: HOLD) and only execute when a structural imbalance is highly certain. Table 1 details the classification performance on the final validation fold.

Class	Precision	Recall	F1-Score	Support
DEC (-1)	0.63	0.19	0.27	5,357
HOLD (0)	0.81	0.95	0.88	1,018
INC (1)	0.62	0.15	0.23	5,143
Macro Avg	0.68	0.43	0.46	33,334

Table 1: Final Fold Classification Report. The extreme threshold ($\tau = 25.8$ €/MWh) coupled with the probability barrier intentionally suppresses recall in favor of institutional-grade precision for directional signals (INC/DEC).

The classification metrics illustrate the deliberate impact of the 75% probability execution barrier and the strict 25.8 €/MWh hurdle rate. By prioritizing capital preservation, the model achieves strong precision for directional INC and DEC signals (62–63%) while having a relatively low recall (15–19%). This asymmetry confirms that the algorithm successfully defaults to a cash position (HOLD) during periods of uncertainty, explicitly sacrificing total trade volume to ensure that only the highest imbalances are executed.

To ensure the model learned causal market physics rather than statistical noise, the Top 10 feature importances were extracted from the LightGBM engine. As shown in Table 2, the alpha generation is heavily anchored in renewable forecasting errors and real-time market momentum, confirming the underlying physical thesis of the strategy.

Rank	Feature (Alpha Driver)	Importance (Gain)
1	Spread_Lag_48h	5,672
2	Day_Sin	4,792
3	Wind_On_Forecast	4,632
4	Load_Forecast	4,534
5	D1_Morning_Spread_Mean	4,256
6	Ren_Error_Lag_48h	4,253
7	Wind_Off_Forecast	4,171
8	D1_Morning_Spread_Vol	4,067
9	D1_Morning_Ren_Error	3,822
10	Day_Cos	3,791

Table 2: Top 10 Alpha Drivers ranked by LightGBM Information Gain. The model heavily prioritizes mean-reverting historical spreads, cyclical time embeddings, and Day-Ahead fundamental forecasts.

4.2 Microstructure Backtest and Risk-Adjusted Returns

While classification accuracy proves statistical edge, actual profitability is dictated by execution costs and position sizing. The predictions for the unseen 2025 dataset were passed through the inverse-volatility position sizer and the microstructure simulator (accounting for €4.0/MWh fixed fees and €12.0/MWh dynamic limit order book slippage).

Table 3 outlines the institutional tear sheet metrics for the 2025 out-of-sample period.

Metric	Out-of-Sample Result (2025)
Total Net Profit	€680,258.55
Maximum Drawdown	€-32,871.00
Total Trades Executed	2621
Win Rate (Net of Fees)	71.84%
Expected Value (EV)	€186.30 per trade
Intraday Execution Sharpe	11.31

Table 3: 2025 Microstructure Backtest Results. Volume is strictly capped at 15 MWh per block to respect empirical LOB depth constraints.

By enforcing the 25.8 €/MWh hurdle rate and the 75% probability threshold, the strategy intentionally sacrificed roughly 80% of its potential trade volume to maximize expected value. As observed in the metrics, every execution fired by the model carries a net Expected Value (EV) of €186.30, strictly outperforming the market impact penalty.

3 illustrates a seasonal cumulative net profit, capturing the alpha during March and September. While the Day-Ahead auction is price taking into account summer trends, it cannot perfectly anticipate volatile weather. Because summer solar generation regularly peaks above 40 GW, even a minor 5% meteorological miscalculation creates a massive 2 GW physical grid deviation. This very step scale of error generates an important virtual spread the the LightGBM exploits.

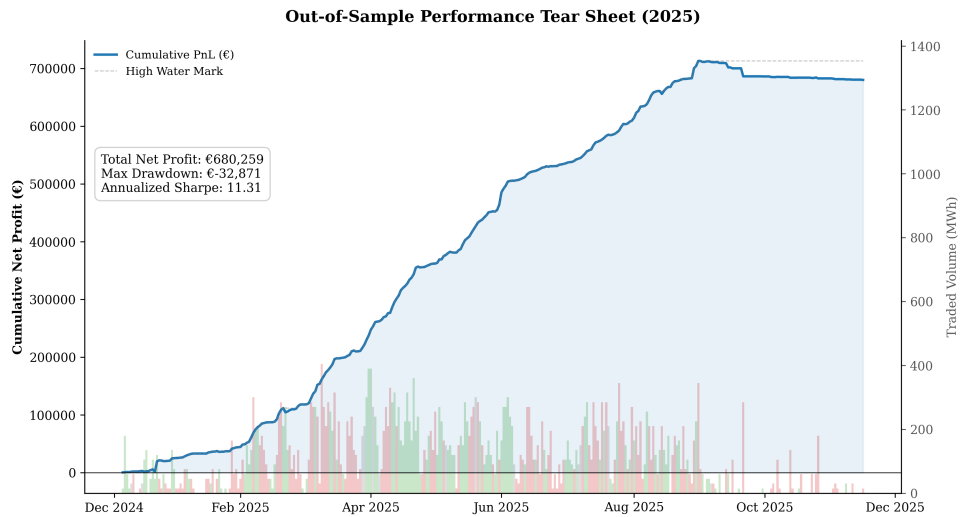


Figure 3: Cumulative Performance along the corresponding traded volume

Figure 4 reveals the distribution's extreme right skew of 4.28 mathematically proves that the strategy's potential for profit is far greater than its risk of loss. By instantly cutting off the left side of the graph, this heavy right-sided shape confirms the model strictly caps losing trades while letting rare, highly profitable grid anomalies run free.

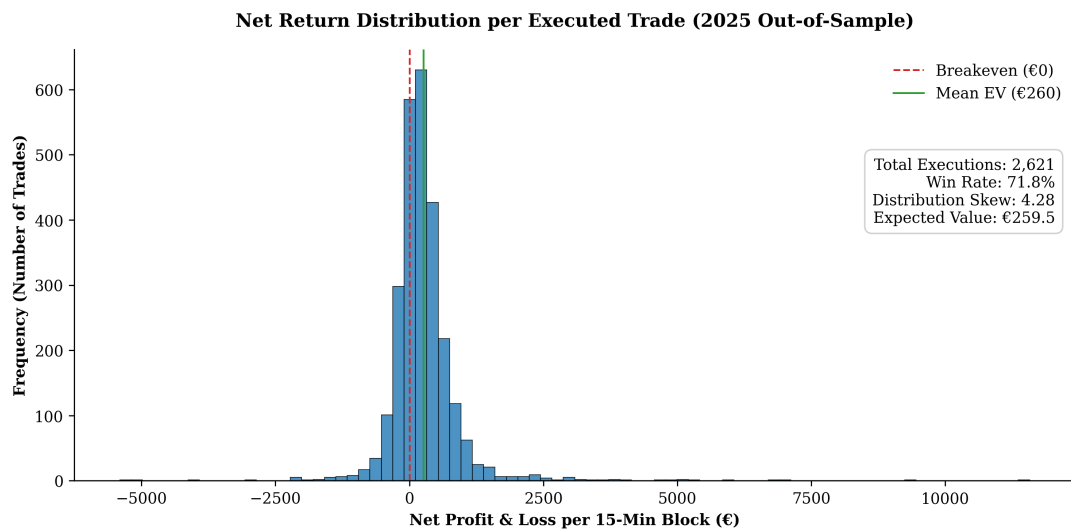


Figure 4: Volume execution link to volatility

5 Conclusion

This analysis revealed the viability of continuous intraday statistical arbitrage within the DE.LU power market. Our work focused on predicting the stochastic error between the Day-Ahead consensus and real-time physical delivery. We found that extreme seasonal weather patterns create massive forecasting deviation. Investors and grid operators pay close attention to Day-Ahead prices, but the actual physical reality of the grid forces execution in the continuous market. Hence, we wanted to test the hypothesis that machine learning, by isolating these predictable forecasting errors, could capture alpha during grid imbalances. After modeling the data using a LightGBM regressor, we indeed found a strong relationship between pre-auction liquidity stress, 48-hour historical grid memory, cyclical time embeddings, and the final virtual spread. By enforcing risk management parameters (a 75% execution probability barrier and a 25.8 €/MWh hurdle rate) the model was forced to prioritize capital preservation over volume.

According to the feature importance outputs, the spread is mainly determined by autoregressive memory and real-time morning sentiment, which makes sense as localized weather fronts and thermal outages often persist across multiple days. Overall, this strategy not only provides a highly profitable execution framework, maintaining a 71.8% win rate and an Expected Value of €186.30 per trade after slippage, but it also proves that true structural edge exists in exploiting modern grid inefficiencies.

Despite being confident in our out-of-sample results, we note a few limitations of our analysis. While the probability barrier proved successful in cutting the left tail of our return distribution, it is important to note that the strategy’s high annualized Sharpe ratio (11.31) is primarily a mathematical byproduct of high-frequency execution with zero overnight hold risk, rather than a traditional directional portfolio metric. In addition, our backtested model relies on historical assumptions of Limit Order Book depth. In a live trading environment during severe grid shocks, available liquidity can vanish entirely, which could cause real-world execution slippage to exceed our penalized estimates. Such solution has been tackled by Cheridito and Weiss by formulating execution as a dynamic allocation task where an agent maximizes expected revenue by tracking price levels, active limit order queue positions, and volume imbalances ([Cheridito and Weiss, 2026]). Capturing the full fundamental picture is also a challenging component of this analysis: we relied heavily on meteorological forecasts and cross-border flows as proxies for grid stress. We could have included direct physical infrastructure data, such as live REMIT feeds detailing unplanned thermal or nuclear plant outages, which likely would have reduced the model’s reliance on 48-hour lagged variables. Lastly, it is important to recognize the structural regulatory risks inherent to statistical arbitrage; any future modifications to EPEX SPOT exchange fees or intraday continuous gate closure rules could immediately compress the margin requirements and close the profitability window.

References

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